

EQ Credit: Daylight

This credit applies to

- ▶ BD+C: New Construction (1-3 points)
- ▶ BD+C: Core & Shell (1-3 points)
- ▶ BD+C: Schools (1-3 points)
- ▶ BD+C: Retail (1-3 points)
- ▶ BD+C: Data Centers (1-3 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-3 points)
- ▶ BD+C: Hospitality (1-3 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To connect building occupants with the outdoors, reinforce circadian rhythms, and reduce the use of electrical lighting by introducing daylight into the space.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Provide manual or automatic (with manual override) glare-control devices for all regularly occupied spaces. For core and shell projects, glare-control devices are not required in the spaces to be fit-out by the tenant(s).

AND

Select one of the following three options.

Option 1. Simulation: Spatial Daylight Autonomy and Annual Sunlight Exposure (1-3 points, 1-2 points Healthcare)

Perform annual computer simulations for spatial daylight autonomy_{300/50%} (sDA_{300/50%}), and annual sunlight exposure_{1000,250} (ASE_{1000,250}) as defined in IES LM-83-12 for each regularly occupied space. Healthcare projects must use each regularly occupied space located in the perimeter area determined under EQ Credit Quality Views. Additionally, calculate the average sDA_{300/50%} value for the total regularly occupied floor area.

For any regularly occupied spaces with ASE_{1000,250} greater than 10%, identify how the space is designed to address glare.

Points are awarded according to Table 1.

Table 1. Points for Option 1

	<i>New Construction, Core and Shell, Schools, Retail, Data Centers, Warehouses and Distribution Centers, Hospitality</i>	Healthcare
The average sDA _{300/50%} value for the regularly occupied floor area is at least 40%	1 point	1 point
The average sDA _{300/50%} value for the regularly occupied floor area is at least 55%	2 points	2 points

The average sDA _{300/50%} value for the regularly occupied floor area is at least 75%	3 points	Exemplary performance
Each regularly occupied space achieves sDA _{300/50%} value of at least 55%	Exemplary performance <i>or 1 additional point if 2 points achieved above.</i>	Exemplary performance

The sDA and ASE calculation grids should be no more than 2 feet (600 millimeters) square and laid out across the regularly occupied area at a work plane height of 30 inches (760 millimeters) above finished floor (unless otherwise defined). Use an hourly time-step analysis based on typical meteorological year data, or an equivalent, for the nearest available weather station. Include any permanent interior obstructions. Moveable furniture and partitions may be excluded.

CS only

If the finishes in the space will not be completed, use the following default surface reflectances: 80% for ceilings, 20% for floors, and 50% for walls. Assume that the entire floor plate, except for the core, will be regularly occupied space.

OR

Option 2. Simulation: Illuminance Calculations (1-3 points, 1-2 points Healthcare)

Perform computer simulations for illuminance at 9 a.m. and 3 p.m. on a clear-sky day at the equinox for each regularly occupied space. Healthcare projects should use the regularly occupied spaces located in the perimeter area determined under EQ Credit Quality Views.

Demonstrate illuminance levels are between 300 lux and 3,000 lux at both 9 a.m. and 3 p.m. Spaces with view-preserving automatic (with manual override) glare-control devices may demonstrate compliance for only the minimum 300 lux illuminance level.

Points are awarded according to Table 2.

Table 2. Points for Option 2

<i>New Construction, Core and Shell, Schools, Retail, Data Centers, Warehouses and Distribution Centers, Hospitality</i>		<i>Healthcare</i>	
<i>Percentage of regularly occupied floor area</i>	<i>Points</i>	<i>Percentage of regularly occupied floor area within perimeter area</i>	<i>Points</i>
55%	1	55%	1
75%	2	75%	2
90%	3	90%	Exemplary performance

Calculate illuminance intensity for sun (direct component) and sky (diffuse component) for clear-sky conditions as follows:

- ▶ Use typical meteorological year data, or an equivalent, for the nearest available weather station.
- ▶ Select one day within 15 days of September 21 and one day within 15 days of March 21 that represent the clearest sky condition.
- ▶ Use the average of the hourly value for the two selected days.

Exclude blinds or shades from the model. Include any permanent interior obstructions. Moveable furniture and partitions may be excluded.

Core and Shell only

Assume the following default surface reflectances if the finishes in the space will not be completed: 80% for ceilings, 20% for floors, and 50% for walls. Assume that the entire floor plate, except for the core, will be regularly occupied space.

OR

Option 3. Measurement (1-3 points, 1-2 points Healthcare)

Measure illuminance in each regularly occupied space. Healthcare projects should use the regularly occupied spaces located in the perimeter area determined under EQ Credit Quality Views.

Achieve illuminance levels between 300 lux and 3,000 lux. Spaces with view-preserving automatic (with manual override) glare-control devices may demonstrate compliance for only the minimum 300 lux illuminance level.

Points are awarded according to Table 3.

Table 3. Points for Option 3

<i>New Construction, Core and Shell, Schools, Retail, Data Centers, Warehouses and Distribution Centers, Hospitality</i>		<i>Healthcare</i>	
<i>Percentage of regularly occupied floor area</i>	<i>Points</i>	<i>Percentage of regularly occupied floor area within perimeter area</i>	
55% at one time in the year	1	55% at one time in the year	1
75% at two times in the year	2	75% at two times in the year	2
90% at two times in the year	3	90% at two times in the year	exemplary performance

With furniture, fixtures, and equipment in place, measure illuminance levels as follows:

- ▶ Measure at appropriate work plane height during any hour between 9 a.m. and 3 p.m.
- ▶ If pursuing one point, take one measurement in any regularly occupied month. If pursuing two points, take two measurements: one measurement in any regularly occupied month, and take a second as indicated in Table 4.
- ▶ For spaces larger than 150 square feet (14 square meters), take measurements on a maximum 10 foot (3 meter) square grid.
- ▶ For spaces 150 square feet (14 square meters) or smaller, take measurements on a maximum 3 foot (900 millimeters) square grid.

Table 4. Timing of measurements for illuminance

<i>If first measurement is taken in ...</i>	<i>take second measurement in ...</i>
January	May-September
February	June-October
March	June-July, November-December
April	August-December
May	September-January
June	October-February
July	November-March
August	December-April
September	December-January, May-June
October	February-June
November	March-July
December	April-August

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The changes include making the three options for daylight credit clearer, more detailed and achievable. For Option 1 of the credit (Simulation: SDA and ASE), the thresholds for SDA 300/50% now have a new lower entry threshold of 40% for 1 point. The stringent 10% ASE threshold is no longer required to be met however spaces exceeding this value are required to identify how they are designed to address resulting glare.

For Option 2 (Simulation: Illuminance Calculations), computer simulations for illuminance need to be performed for each individual regularly occupied space. Spaces with view preserving automatic glare-control devices can now show compliance only for the minimum 300 lux level instead of 300-3000 lux levels. Further, a lower entry threshold of 55% is newly introduced for 1 point to encourage daylight performance at design phase and the highest threshold of 90% compliance earns 3 points.

For Option 3 (Measurement), illuminance measurement needs to be performed for each regularly occupied space. Spaces with view preserving glare-control devices can now show compliance only for 300 lux level. A lower entry threshold for 55% of floor area and 1 yearly time-point measurement is introduced for 1 point and highest compliance threshold for 90% of floor area earns 3 points.

Step-by-Step Guidance

Refer to LEED v4 reference guide with the following modifications:

Option 1. Simulation- Spatial Daylight Autonomy

Step 5. Evaluate compliance for annual sunlight exposure

Record the ASE values for each analysis area however, the ASE value does not need to meet 10% for each analysis area. For the exceeded ASE values, identify how the space is designed to address glare.

Option 2. Simulation- Illuminance Calculation

Step 3. Evaluate illuminance compliance. Follow guidance as per LEED v4 reference guide and record all daylit areas with illuminance levels between 300 lux and 3000 lux, however include spaces that have view preserving automatic glare-control devices, these can demonstrate compliance at 300 lux minimum illuminance level.

Option 3: Measurement

Step 3: Evaluate illuminance compliance. Follow guidance as per LEED v4 reference guide and record all daylit areas with illuminance levels between 300 lux and 3000 lux, however include spaces that have view preserving automatic glare-control devices, these can demonstrate compliance at 300 lux minimum illuminance level.

Further Explanation

Connection with Ongoing Building Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to promote connection of building occupants with outdoors and reinforce circadian rhythms by introducing appropriate amount of daylight into newly constructed spaces via measurement and simulation approaches can contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.